Some Number Sequences I

- Let

$$h_0, h_1, h_2, \ldots, h_n, \ldots \tag{7.1}$$

denote a sequence of numbers. We call h_n the *general term* or *generic term* of the sequence. Two familiar types of sequences are

  - *arithmetic sequences*, in which each term is a constant q more than the previous term,
  - *geometric sequences*, in which each term is a constant multiple q of the previous term.

- In both instances, the sequence is uniquely determined once the initial term h_0 and the constant q are specified.
  - arithmetic sequences

$$h_0, h_0 + q, h_0 + 2q, \ldots, h_0 + nq, \ldots \tag{7.2}$$

  - geometric sequences

$$h_0, qh_0, q^2 h_0, \ldots, q^n h_0, \ldots \tag{7.3}$$



Some Number Sequences II

Example

Arithmetic sequences

1. The sequence of odd positive integers

$$h_0 = 1, q = 2; \qquad 1, 3, 5, \ldots, 1 + 2n, \ldots.$$

2. A constant sequence

$$h_0 = 4, q = 0; \qquad 4, 4, 4, \ldots, 4, \ldots.$$

3. The sequence of nonnegative integers (the counting numbers)

$$h_0 = 0, q = 1; \qquad 0, 1, 2, \ldots, n, \ldots.$$



Some Number Sequences III

Example

Geometric sequences

1. The sequence of nonnegative integral powers of 2

$$h_0 = 1, q = 2; \qquad 1, 2, 2^2, \ldots, 2^n, \ldots.$$

2. A sequence of the counting problem that asks for the number of submultisets of the multiset consisting of $n+1$ different objects whose repetition numbers are given by $4, 2, 2, \ldots, 2$ (n 2's), respectively.

$$h_0 = 5, q = 3; \qquad 5, 5 \times 3, 5 \times 3^2, \ldots, 5 \times 3^n, \ldots.$$



Some Number Sequences IV

- The partial sums for a sequence, $h_0, h_1, h_2, \ldots, h_n, \ldots$, are the sums

$$
\begin{aligned}
s_0 &= h_0 \\
s_1 &= h_0 + h_1 \\
s_2 &= h_0 + h_1 + h_2 \\
&\vdots \\
s_n &= h_0 + h_1 + h_2 + \cdots + h_n = \sum_{k=0}^{n} h_k \\
&\vdots
\end{aligned}
$$

The partial sums form a new sequence $s_0, s_1, s_2, \ldots, s_n, \ldots$ with general term s_n.



Some Number Sequences V

- The partial sums for an arithmetic sequence are

$$s_n = \sum_{k=0}^{n}(h_0 + kq) = (n+1)h_0 + \frac{qn(n+1)}{2}.$$

- The partial sums for a geometric sequence are

$$s_n = \sum_{k=0}^{n} q^k h_0 = \begin{cases} \frac{q^{n+1}-1}{q-1}h_0 & (q \neq 1) \\ (n+1)h_0 & (q = 1). \end{cases}$$

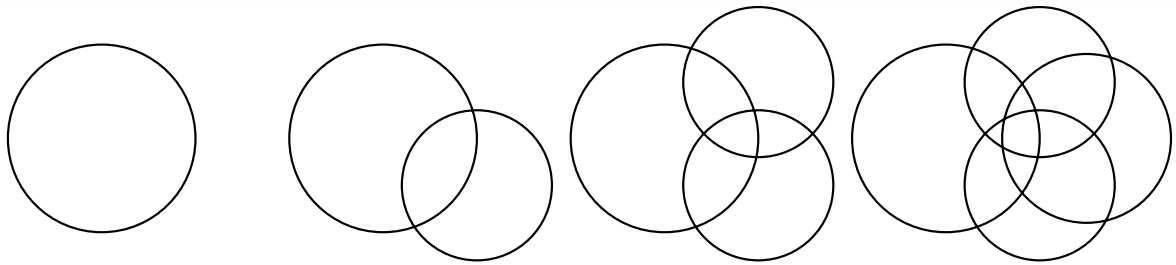


Some Number Sequences VI

- The rules for obtaining the next term in either an arithmetic sequence or geometric sequence are simple instances of recurrence relations.

Example

Determine the number h_n of regions that are created by n mutually overlapping circles in general position in the plane. By mutually overlapping we mean that each two circles intersect in two distinct points (thus nonintersecting or tangent circles are not allowed). By general position, we mean that there do not exist three circles with a common point.



Mutually overlapping circles.

Some Number Sequences VII

Solution

We have $h_0 = 1$, $h_1 = 2$, $h_2 = 4$, and $h_3 = 8$. h_n satisfies the recurrence relation

$$h_n = h_{n-1} + 2(n-1), \qquad (n \geq 2). \tag{7.4}$$

A formula for h_n in terms of the parameter n is obtained as

$$h_n = n^2 - n + 2, \qquad (n \geq 2),$$

which is also valid for $n = 1$, though it does not hold for $n = 0$ ($h_0 = 1$).



Some Number Sequences VIII

The problem posed by Leonardo [Fibonacci]

A newly born pair of rabbits of opposite sexes is placed in an enclosure at the beginning of a year. Beginning with the second month, the female gives birth to a pair of rabbits of opposite sexes each month. Each new pair also gives birth to a pair of rabbits each month starting with their second month. Find the number of pairs of rabbits in the enclosure after one year.

1. In the beginning, there is one pair of rabbits who mature during the first month,
2. so that at the beginning of the second month there is also only one pair of rabbits in the enclosure.
3. During the second month the original pair gives birth to a pair of rabbits,
4. so that there will be two pairs of rabbits at the beginning of the third month.
5. During the third month the newborn pair of rabbits is maturing and only the original pair gives birth.
6. Therefore at the beginning of the fourth month there will be a $2 + 1 = 3$ pairs of rabbits in the enclosure.



Some Number Sequences IX

7. At the beginning of month n the pairs of rabbits in the enclosure can be partitioned into two parts,
8. those present at the beginning of month $n-1$ and those born during month $n-1$.
9. The number of pairs born during month $n-1$ is, because of the 1 month maturation process, the number of pairs that there were at the beginning of month $n-2$.
10. Thus, at the beginning of month n, there are $f_{n-1}+f_{n-2}$ pairs of rabbits.

At the beginning of the month	Pairs of rabbits
1	$f_1 = 1$
2	$f_2 = 1$
3	$f_3 = 2$
4	$f_4 = 3$
$\vdots$	$\vdots$
n	$f_n = f_{n-1} + f_{n-2}$



Some Number Sequences X

- The sequence of numbers $f_0, f_1, f_2, f_3, \ldots$ satisfying the recurrence relation and initial conditions

$$f_0 = 0, \quad f_1 = 1, \quad \text{and} \quad f_n = f_{n-1} + f_{n-2} \quad (n \geq 2) \tag{7.5}$$

  is called the *Fibonacci sequence*, and the terms of the sequence are called *Fibonacci numbers*. The recurrence relation in the above equation is also called the *Fibonacci recurrence*.

Example

The partial sums of the terms of the Fibonacci sequence are

$$s_n = f_0 + f_1 + f_2 + \cdots + f_n = f_{n+2} - 1. \tag{7.6}$$

Example

The Fibonacci number f_n is even if and only if n is divisible by 3.



Some Number Sequences XI

- Consider the Fibonacci recurrence relation in the form

$$f_n - f_{n-1} - f_{n-2} = 0, \quad (n \geq 2) \tag{7.7}$$

  and for the moment, ignore any initial values for f_0 and f_1.
- Look for a solution of the form

$$f_n = q^n,$$

  where q is a nonzero number. Thus, we seek a solution among the familiar geometric sequences with first term equal to $q^0 = 1$.
- We observe that $f_n = q^n$ satisfies the Fibonacci recurrence relation if and only if

$$q^{n-2}(q^2 - q - 1) = 0, \qquad (n = 2, 3, 4, \ldots).$$



Some Number Sequences XII

- Since q is assumed to be different from zero,

$$q_1 = \frac{1+\sqrt{5}}{2}, \qquad q_2 = \frac{1-\sqrt{5}}{2}.$$

- Thus,

$$f_n = \left(\frac{1+\sqrt{5}}{2}\right)^n \quad \text{and} \quad f_n = \left(\frac{1-\sqrt{5}}{2}\right)^n$$

are both solutions of the Fibonacci recurrence relation. Since the Fibonacci recurrence relation is linear and homogeneous, it follows that

$$f_n = c_1\left(\frac{1+\sqrt{5}}{2}\right)^n + c_2\left(\frac{1-\sqrt{5}}{2}\right)^n \tag{7.8}$$

is also a solution of the recurrence relation (7.7) for any choice of constants c_1 and c_2.



Some Number Sequences XIII

- To give a formula for the Fibonacci numbers satisfying initial values with $f_0 = 0$ and $f_1 = 1$, we must have

$$(n = 0)c_1 + c_2 = 0,$$
$$(n = 1)c_1 \left(\frac{1+\sqrt{5}}{2}\right) + c_2 \left(\frac{1-\sqrt{5}}{2}\right) = 1.$$

whose solution is

$$c_1 = \frac{1}{\sqrt{5}}, \qquad c_2 = -\frac{1}{\sqrt{5}}.$$



Some Number Sequences XIV

Theorem

The Fibonacci numbers satisfy the formula

$$f_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n, \quad (n \geq 0). \tag{7.9}$$

Example

Let $g_0, g_1, g_2, \ldots, g_n, \ldots$ be the sequence of numbers satisfying the Fibonacci recurrence relation and the initial conditions as follows,

$$g_n = g_{n-1} + g_{n-2} \qquad (n \geq 2),$$
$$g_0 = 2, \quad g_1 = -1.$$



Some Number Sequences XV

Solution

We would like to determine c_1 and c_2 that satisfy

$$c_1 + c_1 = 2,$$
$$c_1\left(\frac{1+\sqrt{5}}{2}\right) + c_2\left(\frac{1-\sqrt{5}}{2}\right) = -1.$$

and give a formula for g_n as

$$g_n = \frac{\sqrt{5}-2}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n + \frac{\sqrt{5}-2}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n.$$

Example

Determine the number h_n of ways to perfectly cover a 2-by-n board with dominoes.



Some Number Sequences XVI

Solution

Define $h_0 = 1$ (a 2-by-0 board is empty and has exactly one perfect cover, namely the empty cover) and we can conclude that $h_n = h_{n-1} + h_{n-2}$ $(n \geq 2)$.

Example

Determine the number b_n of ways to perfectly cover a 1-by-n board with monominoes and dominoes.



Some Number Sequences XVII

Solution

1. If we take a perfect cover of a 2-by-n board with dominoes and look only at its first row, we see a prefect cover of a 1-by-n board with monominoes and dominoes.
2. Conversely, each perfect cover of a 1-by-n board with monominoes and dominoes can be "extended" uniquely to a perfect cover of a 2-by-n board with dominoes.
3. Thus the numbers of perfect covers of a 1-by-n board with monominoes and dominoes equals the numbers of perfect covers of a 2-by-n board with dominoes.
4. Therefore $b_0, b_1, b_2, \ldots, b_n, \ldots$ is also the Fibonacci sequence with f_0 deleted.



Some Number Sequences XVIII

Theorem

The sums of the binomial coefficients along the diagonals of Pascal's triangle running upward from the left are Fibonacci numbers. More precisely, the nth Fibonacci number f_n satisfies

$$f_n = \binom{n-1}{0} + \binom{n-2}{1} + \binom{n-3}{2} + \cdots + \binom{n-k}{k-1},$$

where $k = \lfloor \frac{n+1}{2} \rfloor$.

Let

$$g_n = \binom{n-1}{0} + \binom{n-2}{1} + \cdots + \binom{n-k}{k-1} \quad (n \geq 0)$$



Some Number Sequences XIX

where $k = \lfloor \frac{n+1}{2} \rfloor$. Since $\binom{n}{p} = 0$ for each integer $p > n$, we can also write

$$g_n = \binom{n-1}{0} + \binom{n-2}{1} + \binom{n-3}{2} + \cdots + \binom{0}{n-1},$$

or

$$g_n = \sum_{k=0}^{n-1} \binom{n-1-k}{k},$$

To prove the theorem, it will suffice to show that g_n satisfies the Fibonacci recurrence relation and has the same initial values as the Fibonacci sequence. We have

$$g_0 = \binom{0}{-1} = 0, \qquad g_1 = \binom{0}{0} = 1.$$



Some Number Sequences XX

Using Pascal's formula, we see that, for each $n \geq 2$,

$$\begin{aligned}
g_{n-1} + g_{n-2} &= \sum_{k=0}^{n-2} \binom{n-2-k}{k} + \sum_{j=0}^{n-3} \binom{n-3-j}{j} \\
&= \binom{n-2}{0} + \sum_{k=1}^{n-2} \binom{n-2-k}{k} + \sum_{k=1}^{n-2} \binom{n-2-k}{k-1} \\
&= \binom{n-2}{0} + \sum_{k=1}^{n-2} \left[\binom{n-2-k}{k} + \binom{n-2-k}{k-1} \right] \\
&= \binom{n-2}{0} + \sum_{k=1}^{n-2} \binom{n-1-k}{k} \\
&= \binom{n-1}{0} + \sum_{k=1}^{n-2} \binom{n-1-k}{k} + \binom{0}{n-1} \\
&= \sum_{k=0}^{n-1} \binom{n-1-k}{k} \\
&= g_n.
\end{aligned}$$

We conclude that $g_0, g_1, g_2, \ldots, g_n, \ldots$ is the Fibonacci sequence and this proves the theorem. □



Linear Homogeneous Recurrence Relations I

- Let $h_0, h_1, h_2, \ldots, h_n, \ldots$ be a sequence of numbers. This sequence is said to satisfy a *linear recurrence relation of order k*, provided that there exist quantities $a_1, a_2, \ldots, a_k$, with $a_k \neq 0$, and a quantity b_n (each of these quantities $a_1, a_2, \ldots, a_k, b_n$ may depend on n) such that

$$h_n = a_1 h_{n-1} + a_2 h_{n-2} + \cdots + a_k h_{n-k} + b_n, \quad (n \geq k). \tag{7.10}$$

Example

The sequence of derangement numbers $D_0, D_1, \ldots, D_n, \ldots$ satisfies the two recurrence relations

$$\begin{aligned} D_n &= (n-1)D_{n-1} + (n-1)D_{n-2}, && (n \geq 2) \\ D_n &= nD_{n-1} + (-1)^n, && (n \geq 1). \end{aligned}$$

The first recurrence relation has order 2 and we have $a_1 = n-1$, $a_2 = n-1$ and $b_n = 0$. The second recurrence relation has order 1 and we have $a_1 = n$ and $b_n = (-1)^n$.

Linear Homogeneous Recurrence Relations II

Example

The Fibonacci sequence $f_0, f_1, f_2, \ldots, f_n, \ldots$ satisfies the recurrence relation

$$f_n = f_{n-1} + f_{n-2}, \qquad (n \geq 2)$$

of order 2 with $a_1 = 1$, $a_2 = 1$, and $b_n = 0$.

Example

The factorial sequence $h_0, h_1, h_2, \ldots, h_n, \ldots$ where $h_n = n!$ satisfies the recurrence relation

$$h_n = nh_{n-1}, \qquad (n \geq 1)$$

of order 1 with $a_1 = n$ and $b_n = 0$.



Linear Homogeneous Recurrence Relations III

Example

The geometric sequence $h_0, h_1, h_2, \ldots, h_n, \ldots$ where $h_n = q^n$ satisfies the recurrence relation

$$h_n = qh_{n-1}, \qquad (n \geq 1)$$

of order 1 with $a_1 = q$ and $b_n = 0$.

- The linear recurrence relation (7.10) is called *homogeneous* provided that b_n is the zero constant and is said to have *constant coefficients* provided that $a_1, a_2, \ldots, a_k$ are constants.



Linear Homogeneous Recurrence Relations IV

- We discuss a special method to solve linear homogeneous recurrence relations with constant coefficients, that is, recurrence relations of the form

$$h_n = a_1 h_{n-1} + a_2 h_{n-2} + \cdots + a_k h_{n-k}, \quad (n \geq k) \tag{7.11}$$

where $a_1, a_2, \ldots, a_k$ are constants and $a_k \neq 0$. It can be rewritten in the form

$$h_n - a_1 h_{n-1} - a_2 h_{n-2} - \cdots - a_k h_{n-k} = 0, \quad (n \geq k) \tag{7.12}$$

Theorem

Let q be a nonzero number. Then $h_n = q^n$ is a solution of the linear homogeneous recurrence relation

$$h_n - a_1 h_{n-1} - a_2 h_{n-2} - \cdots - a_k h_{n-k} = 0, \quad (a_k \neq 0, n \geq k) \tag{7.13}$$

with constant coefficients if and only if q is a root of the polynomial equation

$$x^k - a_1 x^{k-1} - a_2 x^{k-2} - \cdots - a_k = 0. \tag{7.14}$$

Linear Homogeneous Recurrence Relations V

- If the polynomial equation has k distinct roots $q_1, q_2, \ldots, q_k$, then

$$h_n = c_1 q_1^n + c_2 q_2^n + \cdots + c_k q_k^n \tag{7.15}$$

  is the general solution of (7.13) in the following sense: No matter what initial values for $h_0, h_1, \ldots, h_{k-1}$ are given, there are constants $c_1, c_2, \ldots, c_k$ so that (7.15) is the unique sequence which satisfies both the recurrence relation (7.13) and the initial conditions.
- The polynomial equation (7.14) is called the characteristic equations of this recurrence relation (7.13) and its k roots are the characteristic roots.



Linear Homogeneous Recurrence Relations VI

Example

Solve the recurrence relation

$$h_n = 2h_{n-1} + h_{n-2} - 2h_{n-3}, \quad (n \geq 3),$$

subject to the initial values $h_0 = 1$, $h_1 = 2$, and $h_2 = 0$.

The characteristics equation of this recurrence relation is

$$x^3 - 2x^2 - x + 2 = 0.$$

and its three roots are 1, -1, 2.

$$h_n = c_1 1^n + c_2(-1)^n + c_3 2^n = c_1 + c_2(-1)^n + c_3 2^n$$



Linear Homogeneous Recurrence Relations VII

is the general solution. We now want constants c_1, c_2, and c_3 so that

$$\begin{aligned}(n=0)&c_1+c_2+c_3=1,\\(n=1)&c_1-c_2+2c_3=2,\\(n=2)&c_1+c_2+4c_3=0.\end{aligned}$$

The unique solution of this system can be found to be $c_1=2$, $c_2=-\frac{2}{3}$, $c_3=-\frac{1}{3}$. Thus

$$h_n=2-\frac{2}{3}(-1)^n-\frac{1}{3}2^n.$$

Example

Words of length n, using only the three letters a, b, c, are to be transmitted over a communication channel subject to the condition that no word in which two a's appear consecutively is to be transmitted. Determine the number of words allowed by the communication channel.

Linear Homogeneous Recurrence Relations VIII

1. Let h_n denote the number of allowed words of length n.
2. We have $h_0 = 1$ (the empty word) and $h_1 = 3$.
3. Let $n \geq 2$. If the first letter of the word is b or c, then the word can be completed in h_{n-1} ways.
4. If the first letter of the word is a, then the second letter is b or c. If the second letter is b, the word can be completed in h_{n-2} ways. If the second letter is c, the word can also be completed in h_{n-2} ways.
5. Hence h_n satisfies the recurrence relation

$$h_n = 2h_{n-1} + 2h_{n-2}, \qquad (n \geq 2).$$



Linear Homogeneous Recurrence Relations IX

6. The characteristic equation is

$$x^2 - 2x - 2 = 0,$$

and the characteristic roots are

$$q_1 = 1 + \sqrt{3}, \qquad q_2 = 1 - \sqrt{3}.$$

Therefore the general solution is

$$h_n = c_1 \left(1 + \sqrt{3}\right) + c_2 \left(1 - \sqrt{3}\right).$$

We find c_1 and c_2 such that the initial values $h_0 = 1$ and $h_1 = 3$ hold. This leads to

$$(n = 0) c_1 + c_2 = 1,$$
$$(n = 1) c_1 \left(1 + \sqrt{3}\right) + c_2 \left(1 - \sqrt{3}\right) = 3,$$



Linear Homogeneous Recurrence Relations X

which has solution

$$c_1 = \frac{2+\sqrt{3}}{2\sqrt{3}}, \qquad c_2 = \frac{-2+\sqrt{3}}{2\sqrt{3}}.$$

Therefore

$$h_n = \frac{2+\sqrt{3}}{2\sqrt{3}}\left(1+\sqrt{3}\right)^n + \frac{-2+\sqrt{3}}{2\sqrt{3}}\left(1-\sqrt{3}\right)^n \qquad (n \geq 0).$$

is the desired solution.



Linear Homogeneous Recurrence Relations XI

- If the roots $q_1, q_2, \ldots, q_k$ of the characteristic equation are not distinct, then

$$h_n = c_1 q_1^n + c_2 q_2^n + \cdots c_k q_k^n \tag{7.16}$$

is not a general solution of the recurrence relation.



Linear Homogeneous Recurrence Relations XII

Example

The recurrence relation

$$h_n = 4h_{n-1} - 4h_{n-2} \quad (n \geq 2)$$

has characteristic equation $x^2 - 4x + 4 = (x-2)^2 = 0$. Thus, 2 is a twofold characteristic root. In this case, (7.16) becomes

$$h_n = c_1 2^n + c_2 2^n = (c_1 + c_2)2^n = c2^n$$

where $c = c_1 + c_2$ is a new constant. For instance, suppose we prescribe the initial values $h_0 = 1$ and $h_1 = 3$.

$$(n = 0) \quad c = 1, \qquad \text{and} \qquad (n = 1) \quad 2c = 3.$$

But these equations are contradictory. Thus, $h_n = c2^n$ is not a general solution of the given recurrence relation.



Linear Homogeneous Recurrence Relations XIII

Example

Find the general solution of the recurrence relation

$$h_n - 4h_{n-1} + 4h_{n-2} = 0, \quad (n \geq 2)$$

- The characteristic equation is

$$x^2 - 4x + 4 = (x-2)^2 = 0$$

and has roots 2, 2. We know $h_n = 2^n$ is a solution of the recurrence relation. It is easy to show that $h_n = n2^n$ is also a solution. We now conclude that

$$h_n = c_1 2^n + c_2 n 2^n \tag{7.17}$$

is the general solution of the given recurrence relation.



Linear Homogeneous Recurrence Relations XIV

Theorem

Let $q_1, q_2, \ldots, q_t$ be the distinct roots of the following characteristic equation of the linear homogeneous recurrence relation with constant coefficients,

$$h_n = a_1 b_{n-1} + a_2 h_{n-2} + \cdots + a_k h_{n-k}, \quad (a_k \neq 0, n \geq k). \tag{7.18}$$

Then, if q_i is an s_i-fold root of the characteristic equation of (7.18), then the part of the general solution of this recurrent relation corresponding to q_i is

$$\begin{aligned} H_n^{(i)} &= c_1 q_i^n + c_2 n q_i^n + \cdots + c_{s_i} n^{s_i - 1} q_i^n \\ &= (c_1 + c_2 n + \cdots + c_{s_i} n^{s_i - 1}) q_i^n. \end{aligned}$$

The general solution of the recurrence relation is

$$h_n = H_n^{(1)} + H_n^{(2)} + \cdots + H_n^{(t)}. \tag{7.19}$$



Linear Homogeneous Recurrence Relations XV

Example

Solve the recurrence relation

$$h_n = -h_{n-1} + 3h_{n-2} + 5h_{n-3} + 2h_{n-4}, \quad (n \geq 4)$$

subject to the initial values $h_0 = 1, h_1 = 0, h_2 = 1$, and $h_3 = 2$.

- The characteristic equation of this recurrence relation is

$$x^4 + x^3 - 3x^2 - 5x - 2 = 0,$$

which has roots -1, -1, -1, 2. Thus the general solution is

$$h_n = c_1(-1)^n + c_2 n(-1)^n + c_3 n^2(-1)^n + c_4 2^n.$$



Linear Homogeneous Recurrence Relations XVI

is the general solution of the given recurrence relation. c_1, c_2, c_3, and c_4 are determined by the initial conditions so that

$$\begin{array}{lrcrcrcrcl}
(n=0) & c_1 & & & & & & c_4 & = & 1, \\
(n=1) & -c_1 & - & c_2 & - & c_3 & + & 2c_4 & = & 0, \\
(n=2) & c_1 & + & 2c_2 & + & 4c_3 & + & 4c_4 & = & 1, \\
(n=3) & -c_1 & - & 3c_2 & - & 9c_3 & + & 8c_4 & = & 2.
\end{array}$$

The unique solution of this system of equations is $c_1 = \frac{7}{9}$, $c_2 = -\frac{3}{9}$, $c_3 = 0$, $c_4 = \frac{2}{9}$. Thus

$$h_n = \frac{7}{9}(-1)^n - \frac{3}{9}n(-1)^n + \frac{2}{9}2^n.$$



Nonhomogeneous Recurrence Relations I

Example

Tower of Hanoi puzzle – There are three pegs and n circular disks of increasing size on one peg, with the largest disk on the bottom. These disks are to be transferred, on at a time, onto another of the pegs, with the provision that at no time is one allowed to place a larger disk on top of a smaller one. The problem is to determine the number of moves necessary for the transfer.

Let h_n be the number of moves required to transfer n disks. One verifies that $h_0 = 0$, $h_1 = 1$ and $h_2 = 3$. To transfer n disks to another peg we must first transfer the top $n-1$ disks to a peg, transfer the largest disk to the vacant peg, and then transfer the $n-1$ disks to the peg which now contains the largest disk. Thus

$$h_0 = 0, \qquad h_n = 2h_{n-1} + 1, \quad (n \geq 1) \tag{7.20}$$



Nonhomogeneous Recurrence Relations II

To find h_n we iterate (7.20)

$$\begin{aligned} h_n &= 2h_{n-1} + 1 \\ &= 2(2h_{n-2} + 1) + 1 = 2^2 h_{n-2} + 2 + 1 \\ &= 2^2(2h_{n-3} + 1) + 2 + 1 = 2^3 h_{n-3} + 2^2 + 2 + 1 \\ &\vdots \\ &= 2^{n-1}(2h_0 + 1) + 2^{n-2} + \cdots + 2^2 + 2 + 1 \\ &= 2^{n-1} + 2^{n-2} + \cdots + 2^2 + 2 + 1 \end{aligned}$$

Thus

$$h_n = \frac{2^n - 1}{2 - 1} = 2^n - 1, \qquad (n \geq 0). \tag{7.21}$$

In the case of $k \geq 4$ pegs, it is an unsolved problem to determine the minimum number of moves needed to transfer n disks of different sizes on one peg onto a



Nonhomogeneous Recurrence Relations III

different pet, again subject to the rule that a smaller disk is never below a larger disk.

- We now illustrate a technique for solving linear recurrence relation of order 1 with constant coefficients — that is, recurrence relations of the form

$$h_n = ah_{n-1} + b_n, \qquad (n \geq 1). \tag{7.22}$$

Example

Solve

$$h_n = 3h_{n-1} - 4n, \quad (n \geq 1)$$
$$h_0 = 2.$$

We first consider the corresponding homogeneous recurrence relation

$$h_n = 3h_{n-1}, \quad (n \geq 1).$$



Nonhomogeneous Recurrence Relations IV

Its characteristic equation is

$$x - 3 = 0,$$

and hence it has one characteristic root $q = 3$, giving the general solution

$$h_n = c3^n, (n \geq 1). \tag{7.23}$$

We now seek a particular solution of the nonhomogeneous recurrence relation

$$h_n = 3h_{n-1} - 4n, (n \geq 1). \tag{7.24}$$

Try a solution of form

$$h_n = rn + s \tag{7.25}$$

for appropriate numbers r and s. We must have

$$rn + s = 3(r(n-1) + s) - 4n$$



Nonhomogeneous Recurrence Relations V

or, equivalently,

$$rn + s = (3r - 4)n + (-3r + 3s).$$

We obtain

$$\begin{aligned} r &= 3r - 4, \\ s &= -3r + 3s. \end{aligned}$$

Hence, $r = 2$ and $s = 3$, and

$$h_n = 2n + 3 \tag{7.26}$$

satisfies (7.24). We now combine the general solution (7.23) of the homogeneous relation with the particular solution (7.26) of the nonhomogeneous relation to obtain

$$h_n = c3^n + 2n + 3. \tag{7.27}$$



Nonhomogeneous Recurrence Relations VI

This is the general solution for (7.24). The initial condition $h_0 = 2$ gives $c = -1$, and hence

$$h_n = -3^n + 2n + 3 \qquad (n \geq 0)$$

is the solution of the original problem.

- This technique is the discrete analogue of a technique used to solve nonhomogeneous differential equations. It can be summarized as follows.
  1. Find the general solution of the homogeneous relation.
  2. Find a particular solution of the nonhomogeneous relation.
  3. Combine the general solution and the particular solution, and determine values of the constants arising in the general solution so that the combined solution satisfies the initial conditions.
- For some nonhomogeneous parts b_n in $h_n = ah_{n-1} + b_n$, there are certain types of particular solutions to try.
  1. If b_n is a polynomial of degree k in n, then look for a particular solution h_n that is also a polynomial of degree k in n. Thus, try
     1. $h_n = r$ (a constant) if $b_n = d$ (a constant),



Nonhomogeneous Recurrence Relations VII

2. $h_n = rn + s$ if $b_n = dn + e$,
3. $h_n = rn^2 + sn + t$ if $b_n = fn^2 + dn + e$.

2. If b_n is an exponential, then look for a particular solution that is also an exponential. Thus, try

$$h_n = pd^n \quad \text{if } b_n = d^n.$$

Example

Solve

$$h_n = 2h_{n-1} + 3^n, \quad (n \geq 1),$$
$$h_0 = 2.$$



Nonhomogeneous Recurrence Relations VIII

Solution

The general solution is

$$h_n = c2^n + 3 \cdot 3^n.$$

Apply the initial condition $h_0 = 2$, the solution is given as

$$h_n = -2^n + 3^{n+1}, \quad (n \geq 0).$$

- The method just discussed for solving the recurrence relations (7.22) fails in general in case that $a = 1$. In this case, the recurrence becomes

$$h_n = h_{n-1} + b_n, \qquad (n \geq 1) \tag{7.28}$$

and iteration yields

$$h_n = h_0 + b_1 + b_2 + \cdots + b_n.$$



Nonhomogeneous Recurrence Relations IX

Thus, solving (7.28) is the same as summing the series

$$b_1 + b_2 + \cdots + b_n.$$

Example

Solve

$$h_n = h_{n-1} + n^3, \quad (n \geq 1),$$
$$h_0 = 0.$$

Solution

$$h_n = 0^3 + 1^3 + 2^3 + \cdots + n^3 = \frac{(n+1)^2(n+2)^2}{4}, \quad (n \geq 0).$$



Nonhomogeneous Recurrence Relations X

Example

Solve

$$h_n = 3h_{n-1} + 3^n, \quad (n \geq 1),$$
$$h_0 = 2.$$

Try

$$h_n = p3^n$$

as a particular solution. We get

$$p = p + 1,$$

an impossibility. So instead we try

$$h_n = pn3^n$$



Nonhomogeneous Recurrence Relations XI

We get

$$pn3^n = 3p(n-1)3^{n-1} + 3^n,$$

which, after cancelation, gives $p = 1$. Thus, the general solution is

$$h_n = c3^n + n3^n.$$

The solution with the initial condition $h_0 = 2$ is given as

$$h_n = 2 \cdot 3^n + n3^n = (2+n)3^n, \quad (n \geq 0).$$



Generating Functions I

- Let $h_0, h_1, h_2, \ldots, h_n, \ldots$ be an infinite sequence of numbers. Its *generating function* is defined to be the infinite series

$$g(x) = h_0 + h_1 x + h_2 x^2 + \cdots + h_n x^n + \cdots .$$

The coefficient of x^n in $g(x)$ is the nth term h_n of the sequence; thus, x^n acts as a placeholder for h_n.

Sequence	Generating function
$1, 1, 1, \ldots, 1, \ldots$	$\frac{1}{1-x}$
$\binom{m}{0}, \binom{m}{1}, \binom{m}{2}, \cdots, \binom{m}{m}$	$(1+x)^m$
$\binom{\alpha}{0}, \binom{\alpha}{1}, \binom{\alpha}{2}, \ldots, \binom{\alpha}{n}, \cdots$	$(1+x)^\alpha$ (α is a real number)
$h_n = \binom{n+k-1}{n}$, $(n \geq 0)$	$\frac{1}{(1-x)^k}$



Generating Functions II

- It is instructive to recall the derivation of the last formula. We have

$$\begin{aligned}\frac{1}{(1-x)^k} &= \underbrace{\frac{1}{1-x}\times\frac{1}{1-x}\times\cdots\times\frac{1}{1-x}}_{k} \\ &= \underbrace{(1+x+x^2+\cdots)(1+x+x^2+\cdots)\cdots(1+x+x^2+\cdots)}_{k} \\ &= \left(\sum_{e_1=0}^{\infty}x^{e_1}\right)\left(\sum_{e_2=0}^{\infty}x^{e_2}\right)\cdots\left(\sum_{e_k=0}^{\infty}x^{e_k}\right). \qquad (7.29)\end{aligned}$$

x^{e_1} is a typical term of the first factor, x^{e_2} is a typical term of the second factor, ..., x^{e_k} is a typical term of the kth factor. Multiplying these typical terms, we get $x^n = x^{e_1}x^{e_2}\cdots x^{e_k}$, provided that

$$e_1 + e_2 + \cdots + e_k = n. \qquad (7.30)$$



Generating Functions III

Thus, the coefficient of x^n in (7.29) equals the number of nonnegative integral solutions of (7.30), and this number we know to be $\binom{n+k-1}{n}$.

Example

For what sequence is

$$(1+x+x^2+x^3+x^4+x^5)(1+x+x^2)(1+x+x^2+x^3+x^4)$$

the generating function?

Solution

The coefficient of x^n in the product is the number h_n of integral solutions of

$$e_1+e_2+e_3=n$$

in which $0 \le e_1 \le 5$, $0 \le e_2 \le 2$, and $0 \le e_3 \le 4$. Note that $h_n = 0$ if $n > 5+2+4 = 11$.

Generating Functions IV

Example

Determine the generating function for the number of n-combinations of apples, bananas, oranges, and pears where, in each n-combination, the number of apples is even, the number of bananas is odd, the number of oranges is between 0 and 4, and there is at least one pear.

Solution

The problem is equivalent to finding the number h_n of nonnegative integral solutions of

$$e_1 + e_2 + e_3 + e_4 = n$$

where e_1 is even (counting the number of apples), e_2 is odd (counting the number of bananas), $0 \leq e_3 \leq 4$ (counting the number of oranges), and $e_4 \geq 1$ (counting the number of pears). We create one factor for each type of



Generating Functions V

fruit, where the exponents are the allowable numbers in the n-combinations for that type of fruit

$$g(x) = (1 + x^2 + x^4 + \cdots)(x + x^3 + x^5 + \cdots)$$
$$(1 + x + x^2 + x^3 + x^4)(x + x^2 + x^3 + \cdots).$$

Notice that

$$1 + x^2 + x^4 + \cdots = \frac{1}{1 - x^2}$$
$$x + x^3 + x^5 + \cdots = \frac{x}{1 - x^2}$$
$$1 + x + x^2 + x^3 + x^4 = \frac{1 - x^5}{1 - x}$$
$$x + x^2 + x^3 + \cdots = \frac{x}{1 - x}.$$



Generating Functions VI

Thus,

$$g(x) = \frac{1}{1-x^2}\frac{x}{1-x^2}\frac{1-x^5}{1-x}\frac{x}{1-x}$$
$$= \frac{x^2(1-x^5)}{(1-x^2)^2(1-x)^2}.$$

Therefore, the coefficients in the Taylor series for this rational function count the number of combinations of the type considered!

Example

Find the number h_n of bags of fruit that can be made out of apples, bananas, oranges, and pears where, in each bag, the number of apples is even, the number of bananas is multiple of 5, the number of oranges is at most 4, and the number of pears is 0 or 1.

n	0	1	2	3	4	5	6
apples	0	0 0	0 0 2	0 0 2 2	0 0 2 2 4	0 2 2 4 4 0	2 2 4 4 0 0 6
bananas	0	0 0	0 0 0	0 0 0 0	0 0 0 0 0	0 0 0 0 0 5	0 0 0 0 5 5 0
oranges	0	0 1	1 2 0	2 3 0 1	3 4 1 2 0	4 2 3 0 1 0	3 4 1 2 0 1 0
pears	0	1 0	1 0 0	1 0 1 0	1 0 1 0 0	1 1 0 1 0 0	1 0 1 0 1 0 0



Generating Functions VII

The generating function for the number h_n is

$$\begin{aligned} g(x) &= (1+x^2+x^4+\cdots)(1+x^5+x^{10}+\cdots) \\ &\quad (1+x+x^2+x^3+x^4)(1+x) \\ &= \frac{1}{1-x^2}\frac{1}{1-x^5}\frac{1-x^5}{1-x}(1+x) \\ &= \frac{1}{(1-x)^2} \\ &= \sum_{n=0}^{\infty}\binom{n+1}{n}x^n \\ &= \sum_{n=0}^{\infty}(n+1)x^n. \end{aligned}$$

Thus, we see that $h_n = n+1$.



Generating Functions VIII

Example

Determine the generating function for the number h_n of solutions of the equation

$$e_1 + e_2 + \cdots + e_k = n$$

in nonnegative odd integers $e_1, e_2, \ldots, e_k$.



Generating Functions IX

Solution

$$
\begin{aligned}
g(x) &= \underbrace{(x + x^3 + x^5 + \cdots) \cdots (x + x^3 + x^5 + \cdots)}_{k} \\
&= x(1 + x^2 + x^4 + \cdots) \cdots x(1 + x^2 + x^4 + \cdots) \\
&= \frac{x}{1 - x^2} \cdots \frac{x}{1 - x^2} \\
&= \frac{x^k}{(1 - x^2)^k}.
\end{aligned}
$$



Generating Functions X

Example

Let h_n denote the number of nonnegative integral solutions of the equations of the equation

$$3e_1 + 4e_2 + 2e_3 + 5e_4 = n.$$

Find the generating function $g(x)$ for $h_0, h_1, h_2, \ldots, h_n, \ldots$.



Generating Functions XI

Solution

We introduce a change of variable by letting $f_1 = 3e_1$, $f_2 = 4e_2$, $f_3 = 2e_3$, and $f_4 = 5e_4$. Then h_n is also equals the number of nonnegative integral solutions of

$$f_1 + f_2 + f_3 + f_4 = n,$$

where f_1 is a multiple of 3, f_2 is a multiple of 4, f_3 is even, f_4 is a multiple of 5. Hence,

$$\begin{aligned} g(x) =& (1 + x^3 + x^6 + \cdots)(1 + x^4 + x^8 + \cdots) \\ & (1 + x^2 + x^4 + \cdots)(1 + x^5 + x^{10} + \cdots) \\ =& \frac{1}{1 - x^3} \frac{1}{1 - x^4} \frac{1}{1 - x^2} \frac{1}{1 - x^5}. \end{aligned}$$



Generating Functions XII

Example

There is available an unlimited number of pennies, nickels, dimes, quarters, and half-dollar pieces. Determine the generating function $g(x)$ for the number h_n of ways of making n cents with these pieces.

Solution

The number of h_n equals the number of nonnegative integral solutions of the equation

$$e_1 + 5e_2 + 10e_3 + 25e_4 + 50e_5 = n.$$

The generating function is

$$g(x) = \frac{1}{1-x}\frac{1}{1-x^5}\frac{1}{1-x^{10}}\frac{1}{1-x^{25}}\frac{1}{1-x^{50}}.$$



Recurrences and Generating Functions I

- The formula deduced from Newton's binomial theorem

$$\frac{1}{(1-rx)^n} = \sum_{k=0}^{\infty} \binom{n+k-1}{k} r^k x^k, \qquad (|x| < 1/|r|). \tag{7.31}$$

Example

Determine the generating function for the sequence of squares

$$0, 1, 4, \ldots, n^2, \ldots.$$

Solution

By (7.31), with $n = 2$ and $r = 1$,

$$\frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \cdots + nx^{n-1} + \cdots,$$



Recurrences and Generating Functions II

and hence,

$$\frac{x}{(1-x)^2} = x + 2x^2 + 3x^3 + \cdots + nx^n + \cdots .$$

Differentiating, we get

$$\frac{1+x}{(1-x)^3} = 1 + 2^2x + 3^2x^2 + \cdots + n^2x^{n-1} + \cdots .$$

Multiplying by x, we obtain

$$\frac{x(1+x)}{(1-x)^3} = x + 2^2x^2 + 3^2x^3 + \cdots + n^2x^n + \cdots .$$

Therefore, $x(1+x)/(1-x)^3$ is the desired generating function.



Recurrences and Generating Functions III

Example

Solve the recurrence relation

$$h_n = 5h_{n-1} - 6h_{n-2}, \quad (n \geq 2),$$

subject to the initial values $h_0 = 1$ and $h_1 = -2$.

Let $g(x) = h_0 + h_1x + h_2x^2 + \cdots + h_nx^n + \cdots$ be the generating function for $h_0, h_1, \ldots, h_n, \ldots$. We then have the following equations,

$$\begin{aligned} g(x) &= h_0 + h_1x + h_2x^2 + \cdots + h_nx^n + \cdots, \\ -5xg(x) &= -5h_0x - 5h_1x^2 - \cdots - 5h_{n-1}x^n + \cdots, \\ 6x^2g(x) &= 6h_0x^2 + \cdots + 6h_{n-2}x^n + \cdots. \end{aligned}$$

Adding these three equations, we obtain

$$\begin{aligned} (1 - 5x + 6x^2)g(x) = h_0 + (h_1 - 5h_0)x + (h_2 - 5h_1 + 6h_0)x^2 + \cdots \\ + (h_n - 5h_{n-1} + 6h_{n-2})x^n + \cdots. \end{aligned}$$



Recurrences and Generating Functions IV

Since $h_n - 5h_{n-1} + 6h_{n-2} = 0$ $(n \geq 2)$, and since $h_0 = 1$ and $h_1 = -2$, we have

$$(1 - 5x + 6x^2)g(x) = h_0 + (h_1 - 5h_0)x = 1 - 7x.$$

Thus,

$$g(x) = \frac{1 - 7x}{1 - 5x + 6x^2}.$$

From this closed formula for the generating function $g(x)$ we would like to be able to determine a formula for h_n. We observe that

$$g(x) = \frac{1 - 7x}{(1 - 3x)(1 - 2x)} = \frac{5}{1 - 2x} - \frac{4}{1 - 3x},$$

$$\frac{1}{1 - 2x} = 1 + 2x + 2^2x^2 + \cdots + 2^nx^n + \cdots,$$

$$\frac{1}{1 - 3x} = 1 + 3x + 3^2x^2 + \cdots + 3^nx^n + \cdots.$$



Recurrences and Generating Functions V

Therefore,

$$\begin{aligned} g(x) &= 5(1 + 2x + 2^2x^2 + \cdots + 2^nx^n + \cdots) \\ &\quad - 4(1 + 3x + 3^2x^2 + \cdots + 3^nx^n + \cdots) \\ &= 1 + (-2)x + (-16)x^2 + \cdots + (5 \cdot 2^n - 4 \cdot 3^n)x^n + \cdots . \end{aligned}$$

and we have $h_n = 5 \cdot 2^n - 4 \cdot 3^n$.

Example

Let $h_0, h_1, h_2, \ldots, h_n, \ldots$ be a sequence of numbers satisfying the recurrence relation

$$h_n + h_{n-1} - 16h_{n-2} + 20h_{n-3} = 0, \quad (n \geq 3),$$

where $h_0 = 0$, $h_1 = 1$ and $h_2 = -1$. Find a general formula for h_n.



Recurrences and Generating Functions VI

Let $g(x) = h_0 + h_1 x + h_2 x^2 + \cdots + h_n x^n + \cdots$ be the generating function for $h_0, h_1, \ldots, h_n, \ldots$. Adding the four equations,

$$\begin{aligned}
g(x) &= h_0 + h_1 x + h_2 x^2 + h_3 x^3 + \cdots + h_n x^n + \cdots, \\
xg(x) &= \quad + h_0 x + h_1 x^2 + h_2 x^3 + \cdots + h_{n-1} x^n + \cdots, \\
-16x^2 g(x) &= \quad - 16h_0 x^2 - 16h_1 x^3 + \cdots - 16h_{n-2} x^n + \cdots, \\
20x^3 g(x) &= \quad + 20h_0 x^3 + \cdots + 20h_{n-3} x^n + \cdots,
\end{aligned}$$

we obtain

$$\begin{aligned}
(1 + x - 16x^2 + 20x^3)g(x) = h_0 &+ (h_1 + h_0)x + (h_2 + h_1 - 16h_0)x^2 \\
&+ (h_3 + h_2 - 16h_1 + 20h_0)x^3 + \cdots \\
&+ (h_n + h_{n-1} - 16h_{n-2} + 20h_{n-3})x^n + \cdots.
\end{aligned}$$

Since $h_n + h_{n-1} - 16h_{n-2} + 20h_{n-3} = 0$ $(n \geq 3)$, and since $h_0 = 0$, $h_1 = 1$, and $h_2 = -1$, we have

$$(1 + x - 16x^2 + 20x^3)g(x) = x.$$



Recurrences and Generating Functions VII

Hence,

$$g(x) = \frac{x}{1 + x - 16x^2 + 20x^3} = -\frac{\frac{2}{49}}{1 - 2x} + \frac{\frac{7}{49}}{(1 - 2x)^2} - \frac{\frac{5}{49}}{1 + 5x}.$$

We observe that

$$\frac{1}{1 - 2x} = \sum_{k=0}^{\infty} 2^k x^k,$$

$$\frac{1}{(1 - 2x)^2} = \sum_{k=0}^{\infty} \binom{k+1}{k} 2^k x^k = \sum_{k=0}^{\infty} (k+1) 2^k x^k,$$

$$\frac{1}{1 + 5x} = \sum_{k=0}^{\infty} (-5)^k x^k.$$



Recurrences and Generating Functions VIII

Consequently,

$$g(x) = -\frac{2}{49}\sum_{k=0}^{\infty} 2^k x^k + \frac{7}{49}\sum_{k=0}^{\infty}(k+1)2^k x^k - \frac{5}{49}\sum_{k=0}^{\infty}(-5)^k x^k$$
$$= \sum_{k=0}^{\infty}\left[-\frac{2}{49}2^k + \frac{7}{49}(k+1)2^k - \frac{5}{49}(-5)^k\right] x^k.$$

and we have

$$h_n = -\frac{2}{49}2^n + \frac{7}{49}(n+1)2^n - \frac{5}{49}(-5)^n, \quad (n = 0, 1, 2, \ldots).$$



Recurrences and Generating Functions IX

- Let the sequence of numbers $h_0, h_1, h_2, \ldots, h_n, \ldots$ be defined by the recurrence relation

$$h_n + a_1 h_{n-1} + \cdots + a_k h_{n-k} = 0, \quad (n \geq k)$$

of order k and with initial values $h_0, h_1, \ldots, h_{k-1}$. a_k is assumed to be different from 0. Let $g(x)$ be the generating function for our sequence. We find that there are polynomials $p(x)$ and $q(x)$ such that

$$g(x) = \frac{p(x)}{q(x)},$$

where $q(x)$ has degree k and $p(x)$ has degree less than k. Indeed, we have

$$\begin{aligned} q(x) &= 1 + a_1 x + a_2 x^2 + \cdots + a_k x^k, \\ p(x) &= h_0 + (h_1 + a_1 h_0)x + (h_2 + a_1 h_1 + a_2 h_0)x^2 \\ &\quad + \cdots + (h_{k-1} + a_1 h_{k-2} + \cdots + a_{k-1} h_0)x^{k-1}. \end{aligned}$$



Recurrences and Generating Functions X

- Conversely, if we are given a polynomial

$$q(x) = b_0 + b_1 x + \cdots + b_k x^k$$

of degree k with $b_0 \neq 0$ and a polynomial

$$p(x) = d_0 + d_1 x + \cdots + d_{k-1} x^{k-1}$$

of degree less than k, then, using partial fractions and (7.31), we can find a power series $h_0 + h_1 x + \cdots + h_n x^n + \cdots$ such that

$$\frac{p(x)}{q(x)} = h_0 + h_1 x + \cdots + h_n x^n + \cdots .$$

We can write the preceding equation in the form

$$d_0 + d_1 x + \cdots + d_{k-1} x^{k-1} = (b_0 + b_1 x + \cdots + b_k x^k) \\ (h_0 + h_1 x + \cdots + h_n x^n + \cdots).$$



Recurrences and Generating Functions XI

Multiplying out the right side and comparing coefficients, we obtain

$$\begin{cases} b_0 h_0 = d_0, \\ b_0 h_1 + b_1 h_0 = d_1, \\ \vdots \\ b_0 h_{k-1} + b_1 h_{k-2} + \cdots + b_{k-1} h_0 = d_{k-1}, \end{cases} \tag{7.32}$$

and

$$b_0 h_n + b_1 h_{n-1} + \cdots + b_n h_{n-k} = 0, \qquad (n \geq k). \tag{7.33}$$

Since $b_0 \neq 0$, equation (7.33) can be written in the form

$$h_n + \frac{b_1}{b_0} h_{n-1} + \cdots + \frac{b_n}{b_0} h_{n-k} = 0, \qquad (n \geq k).$$

This is a linear homogeneous recurrence relation with constant coefficients that is satisfied by $h_0, h_1, \ldots, h_n, \ldots$. The initial values $h_0, h_1, \ldots, h_{k-1}$ can



Recurrences and Generating Functions XII

be determined by solving the triangular system of (7.32), using the fact that $b_0 \neq 0$.



Recurrences and Generating Functions XIII

Theorem

Let $h_0, h_1, h_2, \ldots, h_n, \ldots$ be a sequence of numbers that satisfies the linear homogeneous recurrence relation

$$h_n + c_1 h_{n-1} + \cdots + c_k h_{n-k} = 0, \quad (c_k \neq 0, n \geq k) \tag{7.34}$$

of order k with constant coefficients. Then its generating function $g(x)$ is of the form

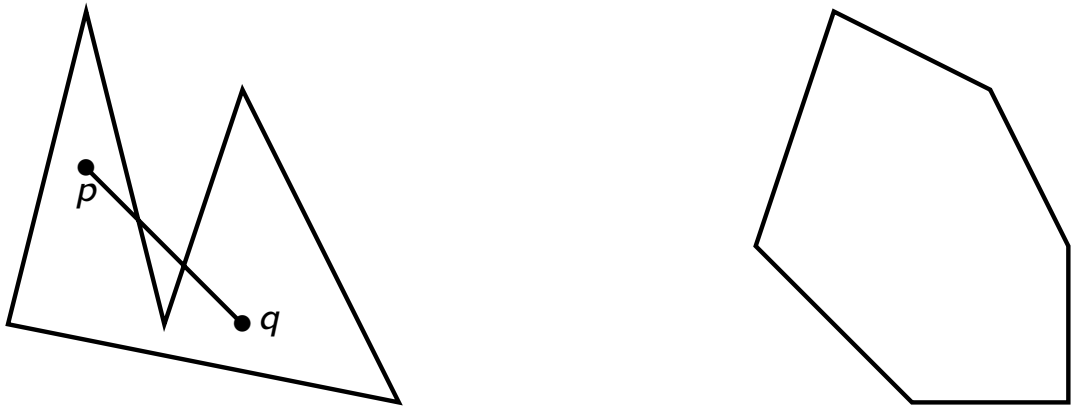
$$g(x) = \frac{p(x)}{q(x)}, \tag{7.35}$$

where $q(x)$ is a polynomial of degree k with a nonzero constant term and $p(x)$ is a polynomial of degree less than k. Conversely, given such polynomials $p(x)$ and $q(x)$, there is a sequence $h_0, h_1, h_2, \ldots, h_n, \ldots$ satisfying a linear homogeneous recurrence relation with constant coefficients of order k of the type (7.34) whose generating function is given by (7.35).



A Geometry Example I

- A set K of points in the plane or in space is said to be *convex*, provided that for any two points p and q in K, all the points on the line segment joining p and q are in K.



(a) (b)

(a) A region which is not convex; (b) A convex polygonal region with six sides.

- The regions are *polygonal regions* provided their boundaries consist of a finite number of line segments, called their *sides*.
- In a polygonal region, the points at which the sides meet are called *corners* (or *vertices*). A *diagonal* is a line segment joining two nonconsecutive corners.
- Let K be a polygonal region with n sides. We can count the number of its diagonals as follows.



A Geometry Example II

1. Each corner is joined by a diagonal to $n-3$ other corners.
2. Counting the number of diagonal at each corner and summing, we get $n(n-3)$.
3. Since each diagonal has two corners, each diagonal is counted twice in this sum.
4. Hence the number of diagonals is $n(n-3)/2$.

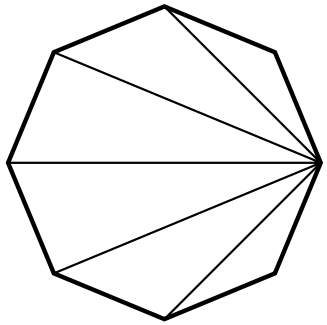
We can arrive at this same number indirectly in the following way.

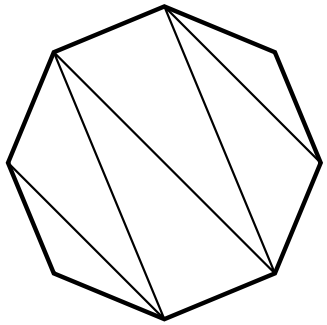
1. There are $\binom{n}{2} = n(n-1)/2$ line segments joining the n corners.
2. Of these, n are sides of the polygonal region. The remaining ones are diagonals.
3. Hence there are $n(n-1)/2 - n = n(n-3)/2$ diagonals.



A Geometry Example III

- Now assume that K is convex. Then each diagonal lies wholly within K. It divides K into one convex polygonal region with k sides and another with $n-k+2$ sides, for some $k = 3, 4, \ldots, n-1$.





Ways of dividing regions into triangular regions by inserting $n-3$ diagonals no two of which intersect in the interior of K.



A Geometry Example IV

Theorem

Let h_n denote the number of ways of dividing a convex polygonal region with $n+1$ sides into triangular regions by inserting diagonals that do not intersect in the interior. Define $h_1 = 1$. Then h_n satisfies the recurrence relation

$$\begin{aligned} h_n &= h_1 h_{n-1} + h_2 h_{n-2} + \cdots + h_{n-1} h_1 \\ &= \sum_{k=1}^{n-1} h_k h_{n-k}, \qquad (n \geq 2). \end{aligned} \tag{7.36}$$

The solution of this recurrence relation is

$$h_n = \frac{1}{n}\binom{2n-2}{n-1}, \qquad (n = 1, 2, 3, \ldots). \tag{7.37}$$

1. We have defined $h_1 = 1$, and we think of a line segment as a polygonal region with two sides and no interior.

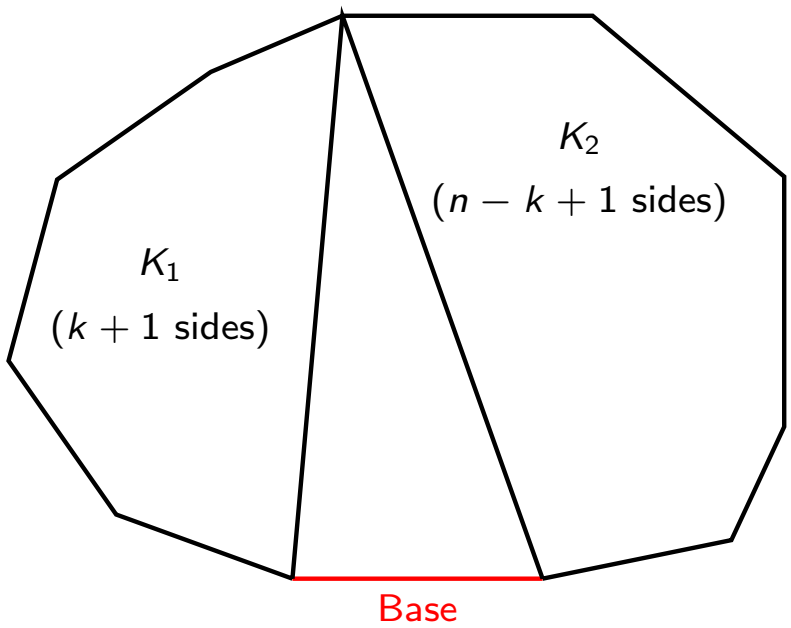


A Geometry Example V

2. We have $h_2 = 1$ since a triangle has no diagonals, and it cannot be further subdivided.
3. The recurrence relations holds for $n = 2, \quad$, since

$$\sum_{k=1}^{2-1} h_k h_{2-k} = \sum_{k=1}^{1} h_k h_{2-k} = h_1 h_1 = 1.$$

4. Now let $n \geq 3$. Consider a convex polygonal region K with $n + 1 \geq 4$ sides.



Polygonal region with $n + 1$ sides.

A Geometry Example VI

5. We distinguish one side of K and call it the *base*. In each division of K into triangle regions, the base is a side of one of the triangular region T, and this triangular region divides the remainder of K into a polygonal region K_1 with $k+1$ sides and a polygonal region K_2 with $n-k+1$ sides, for some $k = 1, 2, \ldots, n-1$ (see Figure).
6. The further subdivision of K is accompanied by dividing K_1 and K_2 into triangular regions by inserting diagonals of K_1 and K_2, respectively, which do not intersect in the interior.
7. Since K_1 has $k+1$ sides, K_1 can be divided into triangular regions in h_k ways. Similarly, K_2 can be divided into triangular regions in h_{n-k} ways.
8. Hence, for a particular choice of the triangular region T containing the base, there are $h_k h_{n-k}$ ways of dividing K into triangular regions by diagonals that do not intersect in the interior.



A Geometry Example VII

9. There are a total of

$$h_n = \sum_{k=1}^{n-1} h_k h_{n-k}$$

ways to divide K into triangular regions in this way. This establishes the recurrence relation (7.36).

10. We now turn to the solution of (7.36) with the initial condition $h_1 = 1$.
11. Let

$$g(x) = h_1 x + h_2 x^2 + \cdots + h_n x^n + \cdots$$

be the generating function for the sequence $h_1, h_2, \ldots, h_n, \ldots$. Multiplying $g(x)$ by itself, we find that

$$\begin{aligned}[g(x)]^2 = h_1^2 x^2 &+ (h_1 h_2 + h_2 h_1) x^3 + (h_1 h_3 + h_2 h_2 + h_3 h_1) x^4 \\ &+ \cdots + (h_1 h_{n-1} + h_2 h_{n-2} + \cdots + h_{n-1} h_1) x^n + \cdots .\end{aligned}$$



A Geometry Example VIII

Using (7.36) and the fact $h_1 = 1$ and $h_2 = 1$, we obtain

$$\begin{aligned}[g(x)]^2 &= h_2x^2 + h_3x^3 + h_4x^4 + \cdots + h_nx^n + \cdots \\ &= g(x) - h_1x = g(x) - x.\end{aligned}$$

12. Thus $g(x)$ satisfies the equation

$$[g(x)]^2 - g(x) + x = 0.$$

13. So $g(x) = g_1(x)$ or $g(x) = g_2(x)$ where

$$g_1(x) = \frac{1 + \sqrt{1 - 4x}}{2}, \qquad g_2(x) = \frac{1 - \sqrt{1 - 4x}}{2}.$$



A Geometry Example IX

14. From the definition, it follows that $g(0) = 0$. Since $g_1(0) = 1$ and $g_2(0) = 0$, we conclude that

$$g(x) = g_2(x) = \frac{1 - \sqrt{1 - 4x}}{2} = \frac{1}{2} - \frac{1}{2}(1 - 4x)^{\frac{1}{2}}.$$

By Newton's binomial theorem

$$(1 + z)^{1/2} = 1 + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n \cdot 2^{2n-1}} \binom{2n-2}{n-1} z^n, \quad (|z| < 1).$$

Replace z by $-4x$, we get

$$g(x) = \sum_{n=1}^{\infty} \frac{1}{n} \binom{2n-2}{n-1} x^n \tag{7.38}$$

and hence

$$h_n = \frac{1}{n} \binom{2n-2}{n-1}, \quad (n \geq 1).$$



Exponential generating functions I

- For sequences whose terms count permutations, it is more useful to consider a generating function with respect to the monomials $\{1, x, \frac{x^2}{2!}, \ldots, \frac{x^n}{n!}, \ldots\}$. These monomials arise in the Taylor series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \cdots .$$

Generating function considered with respect to the monomials $\{1, x, \frac{x^2}{2!}, \ldots, \frac{x^n}{n!}, \ldots\}$ are called *exponential generating function*. The *exponential generating function* for the sequence $h_0, h_1, h_2, \ldots, h_n, \ldots$ is defined to be

$$g^{(e)}(x) = \sum_{n=0}^{\infty} h_n \frac{x^n}{n!} = h_0 + h_1 x + h_2 \frac{x^2}{2!} + \cdots + h_n \frac{x^n}{n!} + \cdots .$$



Exponential generating functions II

Example

Let n be a positive integer. Determine the exponential generating function for the sequence of numbers $P(n,0), P(n,1), \ldots, P(n,n)$. Then exponential generating function is

$$\begin{aligned} g^{(e)}(x) &= P(n,0) + P(n,1)x + P(n,2)\frac{x^2}{2!} + \cdots + P(n,n)\frac{x^n}{n!} \\ &= 1 + nx + \frac{n!}{2!(n-2)!}x^2 + \cdots + \frac{n!}{n!0!}x^n \\ &= (1+x)^n. \end{aligned}$$

Thus, $(1+x)^n$ is both the exponential generating function for the sequence $P(n,0), P(n,1), \ldots, P(n,n)$ and, the ordinary generating function for the sequence

$$\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}.$$



Exponential generating functions III

Example

The exponential generating function for the sequence $1, 1, 1, \ldots, 1, \ldots$ is

$$g^{(e)}(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x.$$

More generally, if a is any real number, the exponential generating function for the sequence $a^0 = 1, a, a^2, \ldots, a^n, \ldots$ is

$$g^{(e)}(x) = \sum_{n=0}^{\infty} a^n \frac{x^n}{n!} = e^{ax}.$$

For a positive integer k, k^n represents the number of n-permutations of a multiset with objects of k different types, each with an infinite repetition number. The exponential function for this sequence of counting number is e^{kx}.



Exponential generating functions IV

Theorem

Let S be the multiset $\{n_1 \cdot a_1, n_2 \cdot a_2, \ldots, n_k \cdot a_k\}$, where $n_1, n_2, \ldots, n_k$ are nonnegative integers. Let h_n be the number of n-permutations of S. Then the exponential generating function $g^{(e)}(x)$ for the sequence $h_0, h_1, h_2, \ldots, h_n, \ldots$ is given by

$$g^{(e)}(x) = f_{n_1}(x) f_{n_2}(x) \cdots f_{n_k}(x), \tag{7.39}$$

where, for $i = 1, 2, \ldots, k$,

$$f_{n_i}(x) = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^{n_i}}{n_i!}. \tag{7.40}$$



Exponential generating functions V

1. Let

$$g^{(e)}(x) = h_0 + h_1 x + h_2 \frac{x^2}{2!} + \cdots + h_n \frac{x^n}{n!} + \cdots$$

be the exponential generating function for $h_0, h_1, h_2, \ldots, h_n, \ldots$. Note $h_n = 0$ for $n > n_1 + n_2 + \cdots + n_k$, so that $g^{(e)}(x)$ is a finite sum.

2. From (7.40), we see that when (7.39) is multiplied out, we get terms of the form

$$\frac{x^{m_1}}{m_1!}\frac{x^{m_2}}{m_2!}\cdots\frac{x^{m_k}}{m_k!} = \frac{x^{m_1+m_2+\cdots+m_k}}{m_1!m_2!\cdots m_k!}, \tag{7.41}$$

where

$$0 \le m_1 \le n_1, 0 \le m_2 \le n_2, \ldots, 0 \le m_k \le n_k.$$

3. Let $n = m_1 + m_2 + \cdots + m_k$. Then the expression in (7.41) can be written as

$$\frac{x^n}{m_1!m_2!\cdots m_k!} = \frac{n!}{m_1!m_2!\cdots m_k!}\frac{x^n}{n!}.$$



Exponential generating functions VI

4. Thus, the coefficient of $x^n/n!$ in (7.39) is

$$\sum \frac{n!}{m_1!m_2!\cdots m_k!}, \tag{7.42}$$

where the summation extends over all integers $m_1, m_2, \ldots, m_k$, with $0 \le m_1 \le n_1, 0 \le m_2 \le n_2, \ldots, 0 \le m_k \le n_k$, and $m_1 + m_2 + \cdots + m_k = n$.

5. A term

$$\frac{n!}{m_1!m_2!\cdots m_k!}$$

in the sum (7.42) equals the number of n-permutations of the submultiset $\{m_1 \cdot e_1, m_2 \cdot e_2, \ldots, m_k \cdot e_k\}$ of S.

6. Since the number of n-permutations of S equals the number of permutations of all such submultisets with $m_1 + m_2 + \cdots + m_k = n$, the number h_n equals the number in (7.42).



Exponential generating functions VII

7. Since this is also the coefficient of $\frac{x^n}{n!}$ in (7.39), we conclude that

$$g^{(e)}(x) = f_{n_1}(x) f_{n_2}(x) \cdots f_{n_k}(x).$$

□

Example

Let h_n denote the number of n-digit numbers with digits 1, 2, or 3, where the number of 1's is even, the number of 2's is at least three, and the number of 3's is at most four. Determine the exponential generating function $g^{(e)}(x)$ for the resulting sequence of numbers $h_0, h_1, h_2, \ldots, h_n, \ldots$.



Exponential generating functions VIII

Solution

$$g^{(e)}(x) = \left(1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right)\left(\frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \cdots\right)$$
$$\left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}\right).$$

Example

Determine the number of ways to color the squares of a 1-by-n chessboard, using the colors, red, white, and blue, if an even number of squares are to be colored red.



Exponential generating functions IX

Solution

$$
\begin{aligned}
g^{(e)}(x) &= \left(1+\frac{x^2}{2!}+\frac{x^4}{4!}+\cdots\right)\left(1+\frac{x}{1!}+\frac{x^2}{2!}+\cdots\right) \\
&\quad \left(1+\frac{x}{1!}+\frac{x^2}{2!}+\cdots\right) \\
&= \frac{1}{2}(e^x+e^{-x})e^x e^x = \frac{1}{2}(e^{3x}+e^x) \\
&= \frac{1}{2}\left(\sum_{n=0}^{\infty} 3^n \frac{x^n}{n!} + \sum_{n=0}^{\infty} \frac{x^n}{n!}\right) \\
&= \frac{1}{2}\sum_{n=0}^{\infty}(3^n+1)\frac{x^n}{n!}.
\end{aligned}
$$

Hence, $h_n = \frac{3^n+1}{2}$.



Exponential generating functions X

Example

Determine the number h_n of n-digit numbers with each digit odd, where the digit 1 and 3 occur an even number of times.



Exponential generating functions XI

Solution

Let $h_0 = 1$. The exponential generating function is

$$\begin{aligned} g^{(e)}(x) &= \left(1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots\right)^2 \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \cdots\right)^3 \\ &= \left(\frac{e^x + e^{-x}}{2}\right)^2 e^{3x} \\ &= \frac{1}{4}\left(e^{5x} + 2e^{3x} + e^x\right) \\ &= \sum_{n=0}^{\infty} \frac{5^n + 2 \cdot 3^n + 1}{4} \frac{x^n}{n!}. \end{aligned}$$

Hence, $h_n = \frac{5^n + 2 \cdot 3^n + 1}{4}$, $(n \geq 0)$.



Exponential generating functions XII

Example

Determine the number h_n of ways to color the squares of a 1-by-n board with the colors red, white, and blue, where the number of red squares is even and there is at least one blue square.



Exponential generating functions XIII

Solution

$$
\begin{aligned}
g^{(e)}(x) &= \left(1+\frac{x^2}{2!}+\frac{x^4}{4!}+\cdots\right)\left(1+\frac{x}{1!}+\frac{x^2}{2!}+\cdots\right)\\
&\quad \left(\frac{x}{1!}+\frac{x^2}{2!}+\cdots\right)\\
&= \frac{1}{2}(e^x+e^{-x})e^x(e^x-1) = \frac{1}{2}(e^{3x}-e^{2x}+e^x-1)\\
&= -\frac{1}{2}+\sum_{n=0}^{\infty}\frac{3^n-2^n+1}{2}\frac{x^n}{n!}.
\end{aligned}
$$

Thus, $h_n = \frac{3^n-2^n+1}{2}$, $(n=1,2,\ldots)$ and $h_0 = 0$.

